APEX ACADEMIC COLLEGE

CAT – Systems Technologies (50 Marks)

QUESTION 1 – MULTIPLE CHOICE (10 marks)

Choose the correct answer.

1.1 The main function of the **CPU** in a computer system is to:
A. Store files permanently
B. Process instructions and perform calculations
C. Display images on the monitor
D. Connect the computer to the internet

1.2 Which of the following is an **input device**?
A. Printer
B. Monitor
C. Keyboard
D. Speaker

1.3 Which type of memory is **volatile**?
A. RAM
B. ROM
C. SSD
D. Hard drive

1.4 A **solid-state drive (SSD)** differs from a hard disk drive (HDD) because:
A. It uses magnetic disks
B. It has no moving parts
C. It is only used in servers
D. It cannot store large files

1.5 Which port is typically used to connect a monitor?
A. HDMI
B. USB
C. Ethernet
D. Audio jack

1.6 Which device protects a computer system from power surges?
A. UPS
B. Router
C. Modem
D. Network switch

1.7 A **network interface card (NIC)** is used to:
A. Process graphics
B. Connect a computer to a network
C. Increase RAM
D. Install software

1.8 Which of the following is an **output device**?
A. Scanner
B. Microphone
C. Projector
D. Webcam

1.9 What is the purpose of an **operating system**?
A. To design documents
B. To manage hardware and software resources
C. To connect to WiFi
D. To play videos

1.10 Which storage device is **portable and commonly used for transferring files**?
A. Flash drive
B. CPU
C. GPU
D. Motherboard

QUESTION 2 – MATCHING ITEMS (10 marks)

Match Column A with Column B.

Column A	Column B
2.1 RAM	A. Permanent storage
2.2 HDD	B. Temporary memory
2.3 Motherboard	C. Connects computer components
2.4 GPU	D. Graphics processing
2.5 Router	E. Connects networks

QUESTION 3 – SHORT QUESTIONS (15 marks)

3.1 Explain the difference between **RAM and ROM**. (2)

3.2 Name **two advantages of SSDs over HDDs**. (2)

3.3 What is the function of the **motherboard**? (2)

3.4 Explain what **cloud storage** is. (2)

3.5 Give **two examples of input devices** used in modern computing. (2)

3.6 Explain what **open-source software** means. (2)

3.7 State **three examples of operating systems**. (3)

QUESTION 4 – SCENARIO-BASED QUESTIONS (15 marks)

A new **school computer lab** is being set up with 25 computers.

4.1 The school wants computers that start quickly and run programs faster.

4.1.1 Which storage device should they choose: **HDD or SSD**? (1)

4.1.2 Give **two reasons** for your answer. (2)

4.2 The computers will be connected to the **internet and to each other**.

4.2.1 Name the device used to **connect the network to the internet**. (1)

4.2.2 Name the device used to **connect multiple computers in the lab network**. (1)

4.3 The school wants to protect computers from **power failures and sudden shutdowns**.

4.3.1 Which device should they install? (1)

4.3.2 Explain how it helps the computers. (2)

4.4 Teachers want students to **store assignments online so they can access them anywhere**.

4.4.1 What technology allows this? (1)

4.4.2 Give **two advantages** of this system. (2)

4.5 Some computers in the lab become **slow after installing many programs**.

Explain **two possible reasons** for this slowdown. (2)